HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026 · PHASE 1

GlucoTwin

A patient digital twin that warns of glucose spikes two hours ahead by fusing health records with wearable data — and tells the doctor what would change the outcome.

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THE PROBLEM

136 million Indians have prediabetes. Most of them are still reversible.

101M
living with diabetes

136M
with prediabetes — still modifiable

2×
times a year a doctor sees them

ICMR-INDIAB national survey, *Lancet Diabetes & Endocrinology*, 2023.

THE GAP

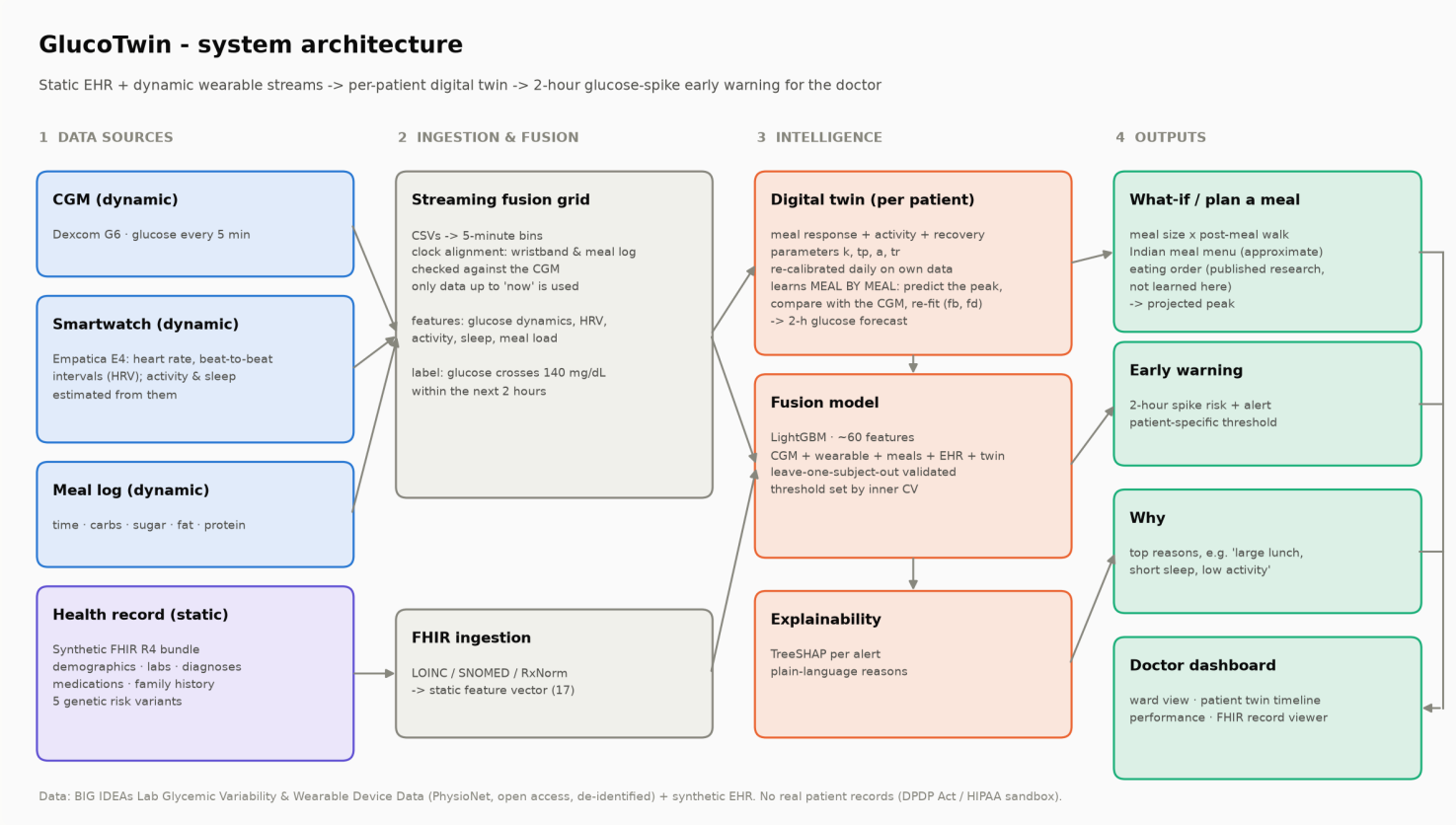
A CGM tells you about a spike. It just tells you too late.

- A doctor sees the patient **twice a year**, and gets one HbA1c — an average over three months.
- A continuous glucose monitor reports the spike **once it is already happening**. By then they have already eaten.
- A simple CGM trend alarm does warn early — at **6.8 false alarms per day**. Nobody keeps listening to that.

Can we warn two hours ahead — and say what to do about it?

THE SYSTEM

Static record + live sensors → a twin per patient



DATA PROVENANCE

What is real, and what is not

STREAM	SOURCE	REAL OR SYNTHETIC
CGM, heart rate, HRV, food logs	BIG IDEAs Lab, PhysioNet (open access)	**Real**, de-identified
Sex, HbA1c	same, Demographics.csv	**Real**, de-identified
Age, labs, diagnoses, medications, family history, genetics	Generated by `src/ehr.py` as FHIR R4	**Synthetic**
Eating-order effect	Shukla et al., *Diabetes Care* 2015	Published research
Indian meal menu	Typical servings	Approximate

16 participants · 8–10 days each · 38,115 five-minute rows · 22% followed by a spike. No real patient record is used anywhere (DPDP Act / HIPAA sandbox).

THE UNGLAMOROUS PART

Three devices. Three clocks.

- **Wristband vs CGM.** Heart rate and glucose are both quiet at night. The shift that lines up the two 24-hour patterns is the offset. Here: **0 hours** — they agreed.
- **Meal log vs CGM.** Glucose should rise 40–80 min after carbs. **Participant 016 was logging meals 105 minutes late** — corrected.
- **Coverage. Participant 015's entire food log falls outside their CGM recording** → marked *unavailable*, meaning "unknown", not "never ate".

Tested on a deliberately broken copy: every offset recovered exactly, AUROC back to within **0.01**.

FEATURE DESIGN

A feature at time T may only use data from time T or earlier

- **~60 features** — CGM dynamics, wearables, sleep, meals, and 17 from the health record.
- **Person-relative wearables.** Resting heart rate varies more *between* people than *within* one. Raw values teach a model *who* someone is — which transfers to a new patient as nothing.
- **Right-closed bins.** A bin labelled 12:05 holds data up to and including 12:05, so a prediction "at 12:05" has never seen 12:06.
- Every rolling statistic is past-only — even the resting heart rate behind the activity estimate.

This is the difference between a glucose model that works and one that is cheating without its author realising.

THE TWIN

Four numbers are the patient

- **k** — carb sensitivity: mg/dL per gram of net carbs
- **tp** — minutes to peak after eating
- **a** — how much activity helps them
- **tr** — how fast they recover

glucose = baseline
+ **k** × carbs, peaking at **tp**
− **a** × recent movement
+ leftover, fading over **tr**

Starts at the population average, then **re-fitted every day on that patient's own past data only**. Only its *forecasts* feed the ML model — the four parameters are constant per person and would act as an ID card.

VALIDATION

Every number comes from a model that never saw that patient

- **Leave-one-subject-out.** Train on 15, test on the 16th, repeat 16 times.
- Random row splits put the same person in train and test and **inflate every score**. Don't.
- **Alert thresholds chosen by inner cross-validation on the training patients only** — nothing about the test patient leaks into the model or the alarm.

RESULTS · LEAVE-ONE-SUBJECT-OUT

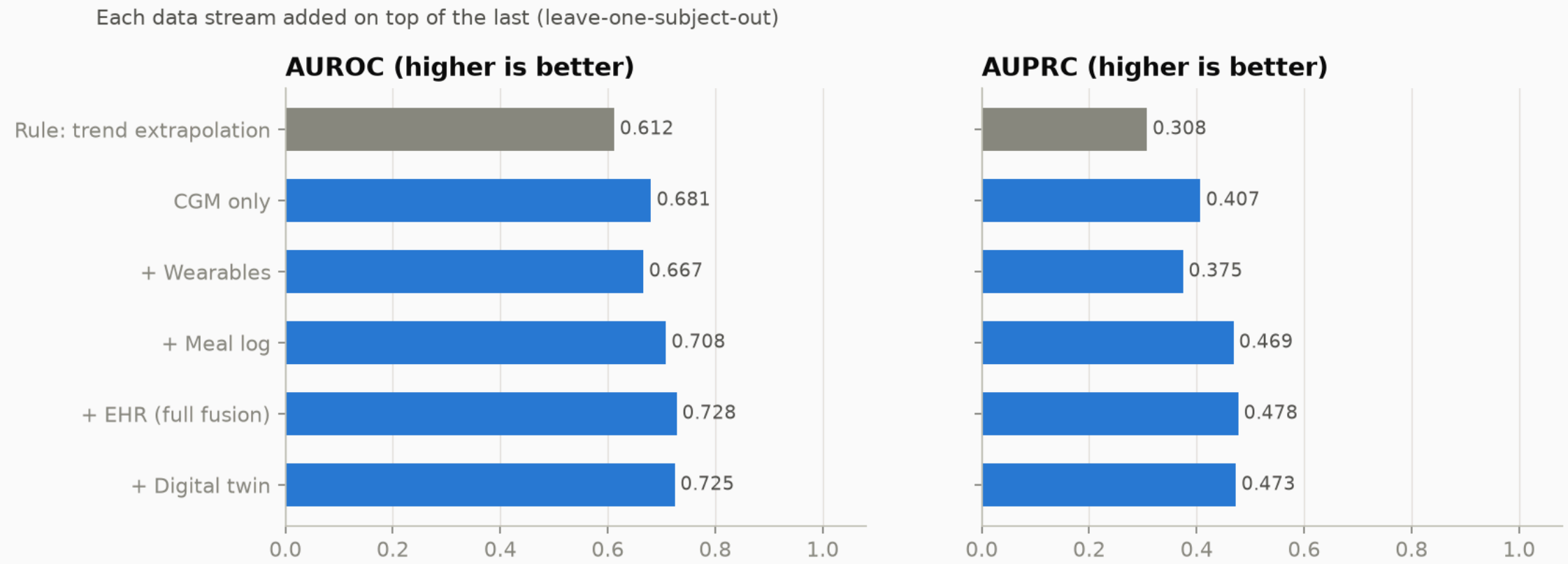
Each row adds one more data stream

DATA USED	AUROC	AUPRC	SPIKES WARNED	MEDIAN LEAD	FALSE ALARMS/DAY
Rule: CGM trend extrapolation	0.612	0.308	99%	75 min	6.8
CGM only	0.681	0.407	91%	10 min	1.7
+ Wearables	0.667	0.375	88%	15 min	2.3
+ Meal log	0.708	0.469	88%	45 min	2.3
+ EHR (full fusion)	**0.728**	**0.478**	**86%**	**45 min**	**2.0**
+ Digital twin forecasts	0.725	0.473	89%	45 min	2.2

AUPRC baseline (random) = **0.22**. Original locked run reported 0.733; re-running on LightGBM 4.7 / pandas 3.0 gives 0.728 — gradient boosting is not bit-reproducible across versions.

RESULTS

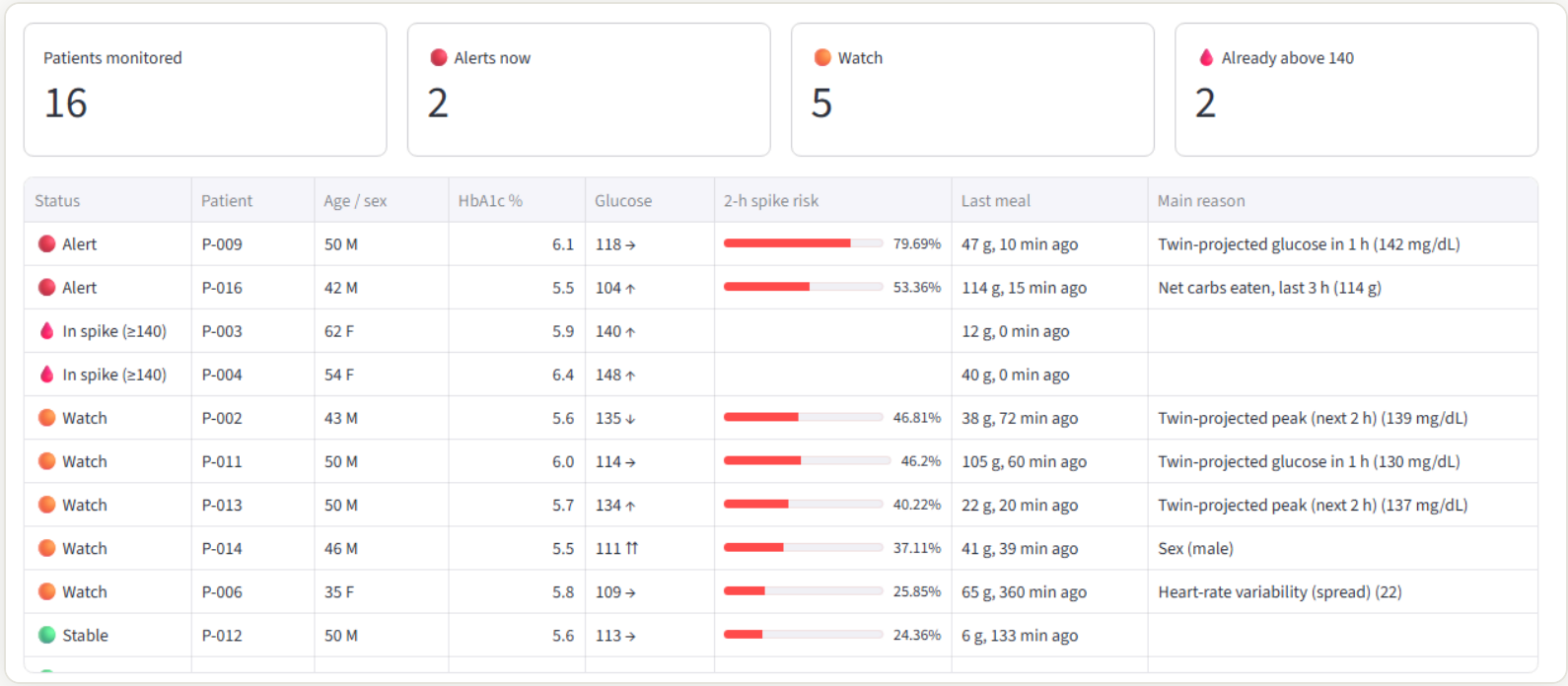
Fusion helps. Wearables alone don't — and I'll say why.



Adding wearables drops the score because the accelerometer files (~13 GB) weren't downloaded — activity is estimated from heart rate, which also rises with stress and after meals. They still contribute inside the full fusion.

THE DOCTOR'S VIEW

Sixteen patients. Two about to spike.



Status	Patient	Age / sex	HbA1c %	Glucose	2-h spike risk	Last meal	Main reason
Alert	P-009	50 M	6.1	118 →	79.69%	47 g, 10 min ago	Twin-projected glucose in 1 h (142 mg/dL)
Alert	P-016	42 M	5.5	104 ↑	53.36%	114 g, 15 min ago	Net carbs eaten, last 3 h (114 g)
In spike (≥140)	P-003	62 F	5.9	140 ↑		12 g, 0 min ago	
In spike (≥140)	P-004	54 F	6.4	148 ↑		40 g, 0 min ago	
Watch	P-002	43 M	5.6	135 ↓	46.81%	38 g, 72 min ago	Twin-projected peak (next 2 h) (139 mg/dL)
Watch	P-011	50 M	6.0	114 →	46.2%	105 g, 60 min ago	Twin-projected glucose in 1 h (130 mg/dL)
Watch	P-013	50 M	5.7	134 ↑	40.22%	22 g, 20 min ago	Twin-projected peak (next 2 h) (137 mg/dL)
Watch	P-014	46 M	5.5	111 ↑↑	37.11%	41 g, 39 min ago	Sex (male)
Watch	P-006	35 F	5.8	109 →	25.85%	65 g, 360 min ago	Heart-rate variability (spread) (22)
Stable	P-012	50 M	5.6	113 →	24.36%	6 g, 133 min ago	

`P-009 · 80%`

"Twin-projected glucose in 1 h — 142 mg/dL."
He ate 47 g of carbs ten minutes ago.

Reasons are **SHAP values** from the model that never saw that patient, turned into plain language.

The two marked **in spike** are already above 140 — no alert, because the job is predicting *onset*, not announcing what the CGM already shows.

NEW IN THIS BUILD

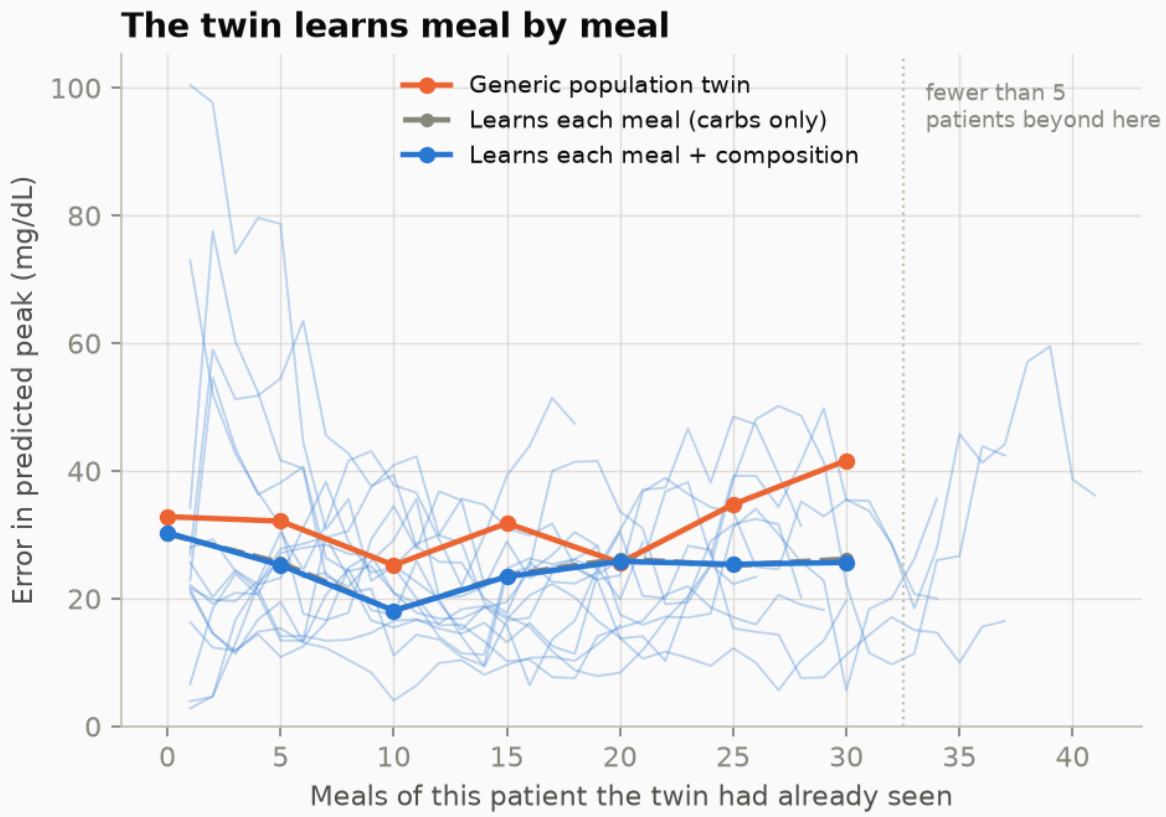
A twin that learns from every meal

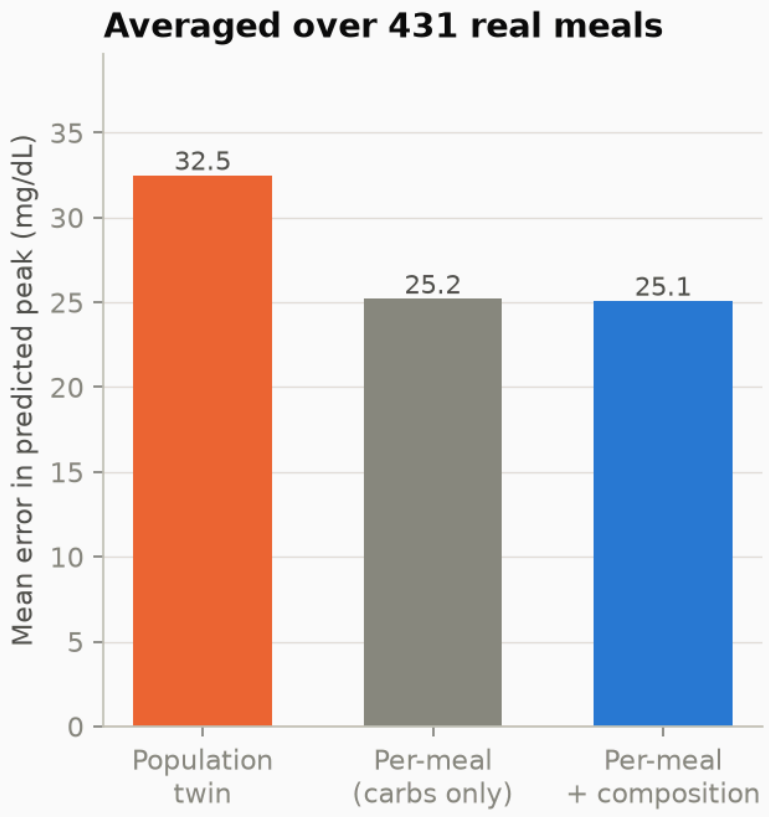
- The daily twin re-fits once a day. But a patient doesn't live in days — **they live in meals**.
- **Before** each meal: predict the peak, from parameters fitted on that person's **earlier meals only**.
- **After** the meal: compare with the real CGM peak.
- **Then** re-fit, so the next meal is predicted by a slightly better twin.

No leakage, and it is **tested, not asserted**: a script deletes future meals and future glucose and checks neither the forecast nor the fitted parameters move. **192 + 53 probes, all unchanged.**

RESULTS · 431 REAL MEALS, 15 PARTICIPANTS

The gap widens the more of you it sees





32.5

mg/dL — generic population twin

25.2

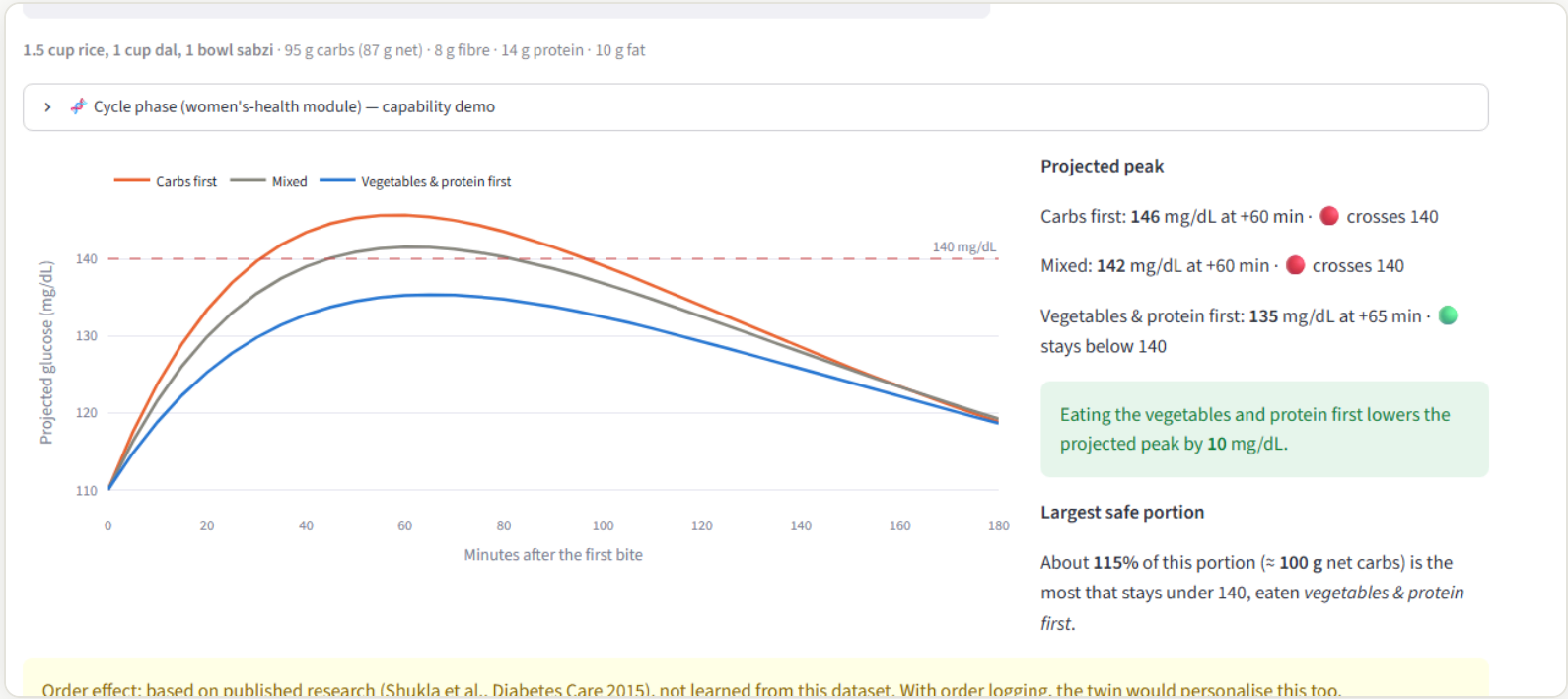
mg/dL — learns each meal (−22%, CI 4.1–10.6, helped 13/15)

+0.12

mg/dL from food composition — CI includes zero. **Did not help.**

PLAN A MEAL · P-005, DAY 3, 16:00

"If she eats the dal before the rice, she won't spike"



The alarm had just called her **24%, green, stable**. Nothing here needed a doctor.

Rice first · 146

crosses 140

Dal & sabzi first · 135

stays under

Her twin: **k = 0.44** mg/dL per g, peak at **65 min** — learned from **21 of her own meals**.

NOT LEARNED FROM THIS DATA

Eating order is a published constant, and the UI says so

- The BIG IDEAs food logs record **what** and **when** — never **in what order**. There is nothing here to learn it from.
- Shukla AP et al., *Diabetes Care* 2015;38(7):e98–e99 — crossover, **n = 11** with type 2 diabetes, reporting **29–37%** lower glucose when vegetables and protein came first.
- I apply **25%** — deliberately below their figure, because eleven people eating a Western meal is not this cohort.
- One named constant: `config.ORDER_PEAK_MULTIPLIER`. Change it or switch it off.

WOMEN'S-HEALTH MODULE

Insulin sensitivity falls in the luteal phase — and this cohort can't show it



The BIG IDEAs study enrolled only post-menopausal females. All 9 women here are post-menopausal. **None has a cycle.**

So it ships as a **labelled capability demo**, behind that warning, defaulting to "Not applicable".

135 → 141

vegetables-first, follicular → luteal. The same advice stops working.

mcPHASES (PhysioNet, 42 women, ~3 months, daily hormone assays) is the dataset that would let the twin *learn* this.

A NEGATIVE RESULT, REPORTED

I re-ran everything on the 9 women alone

DATA USED	AUROC — ALL 16	AUROC — 9 WOMEN
CGM only	0.681	0.607
+ Meal log	**0.708**	**0.667 ← best here**
+ EHR (full fusion)	0.728	0.654
+ Digital twin	0.725	0.655

Full fusion falls to **0.654**, and the health record makes it *worse*. That's a **cohort-size effect, not a finding about women** — nine people, one with no usable food log. It's why the headline keeps all 16.

LIMITATIONS

Arguing against my own project

- **Sixteen people.** Feasibility, not clinical validity.
- **The health record is synthetic** apart from sex and HbA1c — so the EHR result demonstrates the *pipeline*, not real records.
- **Activity and sleep are estimated** from heart rate, not accelerometry or polysomnography.
- **Unlogged meals can't be anticipated.**
- **Eating order and cycle phase are published constants**, labelled on screen every time.
- **Food composition didn't help** — reported as the negative result it is.

WHAT IT DOES SHOW

Not "she will spike" — but "if she eats the dal first, she won't"

0.612→0.728
AUROC, glucose alone → full fusion

6.8→2.0
false alarms per patient-day

−22%
peak error, twin learning meal by meal

Future work: real EHRs through ABDM/FHIR consent flows · a cycle-aware twin learned from mcPHASES · federated on-device training · hypoglycaemia alerts.

Decision support, not a prescription. Not a medical device. Deployment would need prospective validation and regulatory review.

▶ PLAY THE 21-SECOND CLIP NOW

GlucoTwin in twenty-one seconds

- Switch to `brag-output/brag.mp4` and play it full screen, with sound.
- Then come straight back here and move to the live demo.

→ SWITCH TO THE DASHBOARD

Live demo — five minutes

- **Ward overview** · day 3, 20:00 · cohort filter → Women 9
- **Patient twin** → P-005 · day 3 · 16:00 · reasons + timeline
- **Plan a meal** → Rice + dal + sabzi → the three orders → largest safe portion
- **Build your own** / **Describe it in words** → **After the meal** → meal 7
- **Cycle phase** → read the warning → Luteal
- **Model performance** → ablation + per-meal table

THANK YOU

GlucoTwin

Code, data provenance, every result and the leakage test are in the repository.

Varsha B S Acharya · Team Uno · RV University, Bengaluru

github.com/varshabachar/Uno_RVUniversity

Data: BIG IDEAs Lab Glycemic Variability & Wearable Device Data, PhysioNet (ODC-By)